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PAPER

Towards low loss non-volatile phase change materials in mid index waveguides

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Abstract

Photonic integrated circuits currently use platform intrinsic thermo-optic and electro-optic effects to implement dynamic functions such as switching, modulation and other processing. Currently, there is a drive to implement field programmable photonic circuits, a need which is only magnified by new neuromorphic and quantum computing applications. The most promising non-volatile photonic components employ phase change materials such as GST and GSST, which had their origin in electronic memory. However, in the optical domain, these compounds introduce significant losses potentially preventing a large number of applications. Here, we evaluate the use of two newly introduced low loss phase change materials, Sb_2S_3 and Sb_2Se_3, on a silicon nitride photonic platform for future implementation in neuromorphic computing. We focus the study on Mach–Zehnder interferometers that operate at the O and C bands to demonstrate the performance of the system. Our measurements show an insertion loss below 0.04 dB μm^{-1} for Sb_2S_3 and lower than 0.09 dB μm^{-1} for Sb_2Se_3 cladded devices for both amorphous and crystalline phases. The effective refractive index contrast for Sb_2S_3 on SiNx was measured to be 0.05 at 1310 nm and 0.02 at 1550 nm, whereas for Sb_2Se_3, it was 0.03 at 1310 nm and 0.05 at 1550 nm highlighting the performance of the integrated device.

1. Introduction

Photonic integrated circuits (PICs) have recently become an established and powerful technology that supports many applications [1, 2]. PICs are following the trend of integrating electronic integrated circuits, using a range of components to manipulate photons and transfer information such as, on-chip optical waveguides [3], grating couplers (GCs) [4], electro-optic modulators [5–7], photodetectors [8, 9] and lasers on chip [10]. Currently, electronic circuits excel at digital computations, while photonics circuits are demonstrating tremendous progress at transporting and processing analogue and/or digital information at very high data rates [11–14]. Nowadays, PICs are commonly used in fibre-optic communications, but they are also useful in applications in which light has a role, such as chemical, biological or spectroscopic sensors [15, 16], metrology [17], classical and quantum information processing, [11, 18] and artificial neural networks, which is a rapidly emerging area that holds great promise for next generation, low-power pervasive computing technologies [19, 20].

Programmable integrated photonics aim to provide a complementary approach to that based on application-specific photonic integrated circuits, which has been a growing research area over the last few years. The goal is to achieve in the optical domain similar advantages to those brought by field programmable gate arrays over application specific integrated circuits in electronics [21]. Programmable integrated photonics has raised the attention of a range of emerging applications that not only demand flexibility and re-configurability but also enable low-cost, compact and low-power consumption devices. To date, one of the biggest challenges

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faced by programmable silicon photonics is the integration of low-loss non-volatile components in applications such as quantum computing [18, 22, 23], microwave photonics [14], neuromorphic computing [24–26], internet of things (IoT) [27], and machine learning [28].

Silicon nitride (SiNx) is one of the three currently commercially viable photonic platforms [29], the other two being silicon [30] and indium phosphide [31, 32]. SiNx provides an alternative low-cost CMOS compatible platform, and similar to the silicon platform, all fundamental non amplifying photonic components can be implemented [33]. The advantages over Si are fabrication flexibility, low temperature processing (<400 °C), refractive index tunability, high transparency and low temperature sensitivity [34]. As a result, SiNx waveguides have been widely employed for light propagation in the mid infrared, the near infrared and in the visible range of the electromagnetic spectrum [35, 36]. The versatility of the SiNx platform is key in the implementation of complex multi-layer photonic circuitry. Furthermore, the reduced mode confinement compared to Si, allows for compact active regions of programmable devices reducing the energy required to switch them [37, 38].

Phase change materials (PCMs) are being extensively studied for their use in PICs as they offer high refractive index contrast and are non-volatile at the same time. Benefiting from prior art such as rewritable optical media and resistive memories, the mature technology of chalcogenide PCMs are now seen as a promising CMOS compatible route to provide the much needed non-volatile re-configurability in integrated photonics [39]. PCMs have the ability to switch between two states, an amorphous and a crystalline phase with a resulting large optical contrast (allowing also intermediate states of crystallization). They are stable (years at room temperature) [40–42], can be switched between states rapidly (nanoseconds or less) [43, 44], and exhibit high endurance (number of switching cycles) [45, 46]. The fundamental advantage of PCM-based programmable circuits compared to the conventional thermo-optic based approach is that energy is only consumed during the actual switching process. PCM-based devices have been demonstrated in a variety of applications such as switches [47–49], wavelength division multiplexers [50], directional couplers [51], memories [50, 52–54] and neuromorphic devices [55, 56]. Most PCM-based devices to date make use of the well-known pseudo-binary PCM $Ge_2Sb_2Te_5$ (GST). Although it is stablished as a mature technology, GST was designed to offer fast switching (ns) and stability for applications that employ changes in reflectivity through the refractive index contrast, as applied in the CD, DVD, etc, or resistivity differences, applications that are largely not affected by optical loss. On the other hand, PICs that employ GST present strong coupling between amplitude and phase modulation due to the high absorption of the crystalline phase of the material. This coupling severely reduces the potential modulation schemes while it limits systems to a small number of components due to inherent losses. Even though GST-225 offers low losses 0.039 dB μm^{-1} in the amorphous state, the crystalline phase presents losses as high as 2.7 dB μm^{-1} at 1550 nm [57]. Addition of selenium to GST (GSST) presents an alternative solution to non-volatile relatively low-loss PCMs for PICs and different building blocks have been experimentally fabricated and tested [48, 58–60]. These materials still maintain a relatively high loss (extinction coefficient) in order to switch the PCM optically.

A novel family of low-loss PCMs which includes Sb_2S_3 and Sb_2Se_3 has been recently shown [61, 62] and devices taking advantage of their optical properties such as Bragg gratings [63] and ring resonators on silicon-on-insulator (SOI) [64] have been demonstrated. Sb_2S_3 offers a refractive index contrast (Δn) between its states of 0.60 at 1550 nm and 0.58 at 1310 nm, while for Sb_2Se_3, $\Delta n = 0.77$ at 1550 nm and 0.82 at 1310 nm. Both materials present low inherent losses since their extinction coefficient, k is less than 10^{-4} in both phases at 1550 nm and at 1310 nm. The crystallisation temperature has been found to be 290 °C for Sb_2S_3 and 190 °C for Sb_2Se_3. Hence, Sb_2Se_3 is easier to switch due to its lower crystallisation temperature while the lower bandgap energy of 1.48 eV compared to the 1.98 eV of Sb_2S_3 results in a shifted absorption spectrum towards higher wavelengths. As a result, the favourable properties of both materials cover a range of applications.

In this paper, we have experimentally demonstrated for the first time, to the best of our knowledge, Mach–Zehnder interferometers (MZIs) building blocks on a SiNx platform based on recently emerged low loss PCMs, specifically, Sb_2S_3 and Sb_2Se_3 in the O and C-bands. This work shows the advantage of using a SiNx platform to integrate materials to form active individual building blocks and provide a comparison with the more common SOI. We integrate these emerging low-loss PCM materials with a SiNx platform based on PECVD deposition and we demonstrate materials compatibility coupled with high performance in phase modulation. Conceptually, this novel approach uses a CMOS BEOL compatible material using MZIs, which are arguably one of the most mainstream broadband device to obtain dense switch arrays on chip [65]. We demonstrate low losses (<1 dB) and extinction ratios (ERs) as high as 28 and 20 dB in the C and O-band respectively, where PCM (Sb_2S_3) lengths smaller than 40 μm and 14 μm are used to obtain a π phase shift at 1550 and 1310 nm respectively. The length required for obtaining a π shift for Sb_2Se_3 material is 15.5 μm and 21.83 μm for the C and O-band respectively. The effective refractive index contrast and absorption as metrics to assess MZIs operation at two different wavelengths, 1310 nm and 1550 nm have also been evaluated. Our measurements show an insertion loss (IL) below 0.04 dB μm^{-1} for Sb_2S_3 and lower than

0.09 dB μm^{-1} for Sb_2Se_3 cladded devices for both amorphous and crystalline phases. The effective refractive index contrast for Sb_2S_3 on SiNx was measured to be 0.05 at 1310 nm and 0.02 at 1550 nm, whereas for Sb_2Se_3, it was 0.03 at 1310 nm and 0.05 at 1550 nm highlighting the performance of the integrated device.

2. Fabrication and characterization

SiNx MZIs were fabricated using Sb_2S_3 and Sb_2Se_3 as active materials to utilize them as non-volatile phase modulators building blocks. The MZIs building block structures were designed in order to demonstrate a high performance modulation with low loss, deep ER and at the same time test the performance of both Sb_2S_3 and Sb_2Se_3 in the proposed SiNx platform at both 1310 nm and 1550 nm wavelengths. These devices were fabricated on 8" Si wafers with a 2 μm thermally-grown SiO_2 on top of which a 300 nm SiNx layer ($n = 2.0$) was deposited at 350 °C using the NH_3 PECVD process fully detailed in [66]. The test structures were defined on the wafers by means of 248 nm deep-UV (DUV) lithography and the pattern was transferred onto the SiN layer using inductively coupled plasma etching with an etch depth of 300 nm forming ridge structures. The PCMs were deposited through windows defined by the second DUV lithography step. The PCM deposition was performed by RF sputtering as described in [62]. A 10 nm ZnS/SiO_2 capping layer was deposited by sputtering right after the deposition of the PCM. The photoresist was lift-off by sequentially dipping in room temperature acetone and 80 °C NMP with ultrasonic agitation. After rinsing in acetone, IPA and drying with a nitrogen gun a second 10 nm layer of ZnS/SiO_2 was deposited to provide further protection to the exposed PCM sidewalls.

The layout we designed included a set of MZIs with single-mode waveguide cross-sections of 1.2×0.3 μm^2 for 1550 nm (C-band) and 0.9×0.3 μm^2 for 1310 nm (O-band). These MZIs have an optical path length difference of $\Delta L = 60$ μm at 1310 nm and $\Delta L = 40$ μm at 1550 nm between their arms, see figure 1(a). In the longer arm of the MZI, PCM cells with a thickness of 20 nm and different lengths ranging from 2 μm to 125 μm were deposited. To split and combine the light, 3 dB broadband multi-mode interferometers (MMIs) were employed, see inset of figure 1(a). The dimensions of the MMI at $\lambda = 1.31$ μm, were $MMI_L = 41$ μm and $MMI_W = 9$ μm, and respectively for $\lambda = 1.55$ μm, $MMI_L = 64$ μm and $MMI_W = 11$ μm. SEM images of the longer MZI arm were taken with different lengths of the PCM cell. In figure 1(c), a cell length of a 100 μm was captured and in figure 1(d) a zoom in image of the longer arm with a cell length of 20 μm was measured. All the devices included input and output GCs designed to couple the light at either of both wavelengths of interest for characterisation, 1310 or 1550 nm. The GCs designed for 1310 nm consisted of a 10 μm $\times$ 37 μm surface grating with a period of 1038 nm tapered down to a single-mode waveguide width of 900 nm, whereas the ones designed for 1550 nm had a surface grating with the same dimensions and a period of 1238 nm tapered to a single-mode width of 1200 nm. The angle between the optical fibers delivering light and the gratings was selected to be 10° or 15° to the normal to ensure maximum coupling at 1310 or 1550 nm, respectively. Additionally, the layout included separate structures with two GCs connected back-to-back for normalisation purposes. The spectral response for all the devices was characterised using two different tunable laser sources. An Agilent 8163B lightwave multimeter for the response at 1550 nm and a similar Agilent 8164B lightwave measurement system for the response at 1310 nm. The measurements were first performed for the amorphous state of the PCMs and then for their crystalline state. The phase change was induced thermally by heating the chip on a hot plate at the crystallisation temperature for 10 min. In both cases, the polarization of the light was controlled to ensure that only TE modes could propagate through the devices and the measurements were normalized to extract the loss contribution of the GCs.

3. Results and discussion

Firstly, the Sb_2S_3 material is analysed. The bare MZI structure without PCM was characterized and normalized with respect to GCs connected back-to-back. Afterwards, the response of the MZIs for different lengths of the PCM cell were characterized in both amorphous (figures 2(a) and (c)) and crystalline states (figures 2(b) and (d)) at both target wavelengths.

In figure 2(a), the bare MZI presents a free-spectral range (FSR) of 13 nm in the O-band. Only cell lengths of 10, 20 and 30 μm were selected to be plotted in order to not overload the graph with all the cell lengths used in the study. The shift in wavelength ($\Delta\lambda$) produced by the cell lengths was 2.75, 4.38 and 9.47 nm respectively in the amorphous state respect to the bare MZI. In figure 2(b), the crystalline state is shown in the O-band. For the bare structure after crystallization, the FSR maintained the previous value (prior to crystallization), even though a shift has been produced in the dips of the optical spectrum due to the crystallization process to which the chip was submitted. In this case, a shift in wavelength for the cells of length 10, 20 and 30 μm of 7.64, 9.5 and 13.03 nm, respectively, was measured.

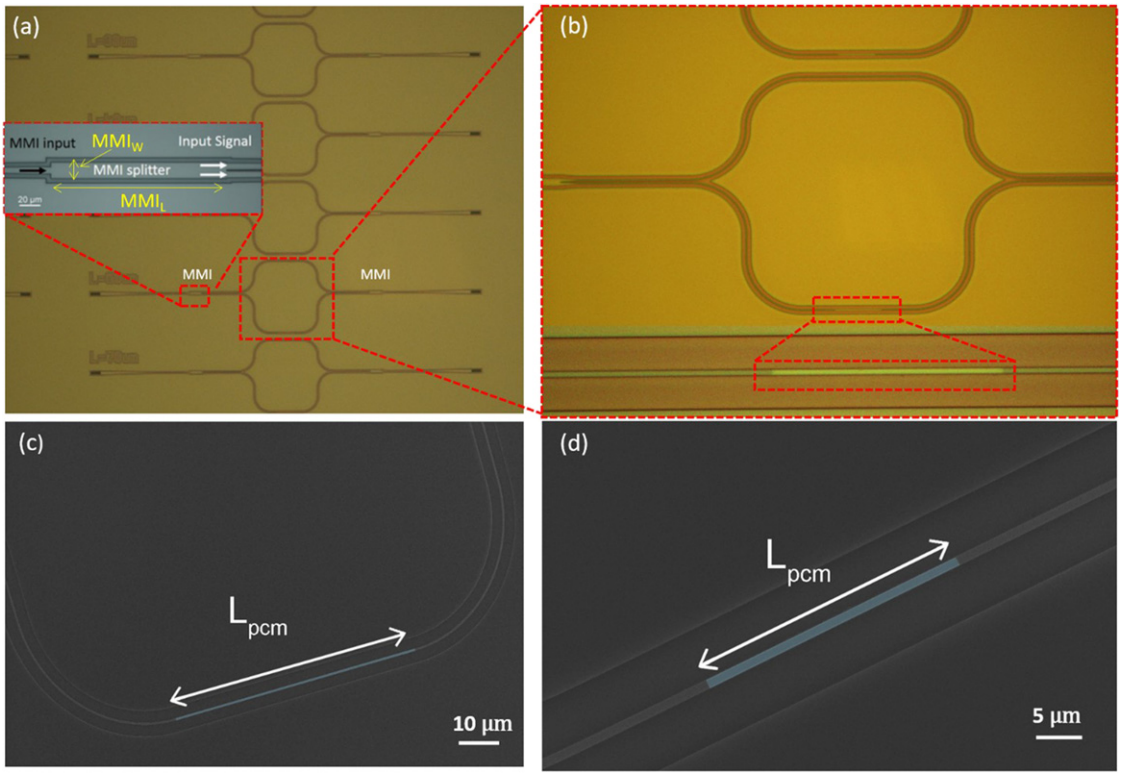

Figure 1. Optical microscope and SEM images of the fabricated structures. (a) MZIs using different cell lengths. (b) Zoom in image of (a) with an inset of the PCM deposited layer. (c) SEM image of the longer arm of the MZI with the cell deposited on top of 100 μm. (d) SEM image of a 20 μm PCM cell with fake color.

In each case, an ER higher than 20 dB is shown for the different MZI structures. Figure 2(c) represents the MZI response using Sb_2S_3 in the amorphous state of the PCM in the C-band. An FSR of 31 nm was extracted, higher than the one measured in the O-band (13 nm), a shift in wavelength for cell lengths of 10, 15 and 20 μm of 9.67, 12.76 and 16.07 nm was experimentally demonstrated. In figure 2(d), the crystalline state of the PCM for the C-band is shown and shifts of 13.5, 19.5 and 22.5 nm were obtained for the three different cell lengths, 10, 15 and 20 μm. ERs as high as 30 dB were experimentally demonstrated in this range of the spectrum.

Afterwards, we characterized the IL introduced by the PCM layer (α) and the modulation in effective refractive index (Δn_{eff}) in its amorphous and crystalline states. The IL and the modulation in phase were extracted from the spectral response of the fabricated MZIs. To characterize the losses of the PCM cell, the ER of each individual MZI with the corresponding cell length was evaluated and consequently the loss coefficient extracted using the equation from reference [67]:

$$\text{ER} = \left(\frac{1 + e^{\frac{-\alpha L_{\text{pcm}}}{2}}}{1 - e^{\frac{-\alpha L_{\text{pcm}}}{2}}} \right)^2 \tag{1}$$

where L_{pcm} is the length of the PCM cell. For the amorphous state of the PCM, the ER was not varying significantly while increasing the cell length, showing a low loss introduced by the PCM in both ranges of the spectrum. The extracted losses in the amorphous state for the wavelengths of interest are lower than 10^{-4} dB μm^{-1}. For the crystalline state, ERs were decreasing with increasing the cell lengths, showing an increment in losses in the crystalline state respect to the amorphous state. The measured IL in the crystalline state, were as low as (0.031±0.003) dB μm^{-1} and (0.023±0.005) dB μm^{-1} at 1310 nm and 1550 nm respectively, see figure 3.

Once the IL of the PCM was characterized, the effective refractive index modulation of the material can be obtained using the equation from reference [68]:

$$\Delta n_{\text{eff}}^{(a-c)} = \frac{\lambda}{\text{FSR}} \cdot \frac{\Delta\lambda^{(a-c)}}{L_{\text{pcm}}} \tag{2}$$

where Δn_{eff} is the variation produced in the effective refractive index when the PCM cell is introduced compared to the bare structure (no PCM), the superscripts a and c refer to the amorphous and crystalline states respectively, FSR is the free spectral range as previously introduced, and $\Delta\lambda$ is the shift in the dip of the

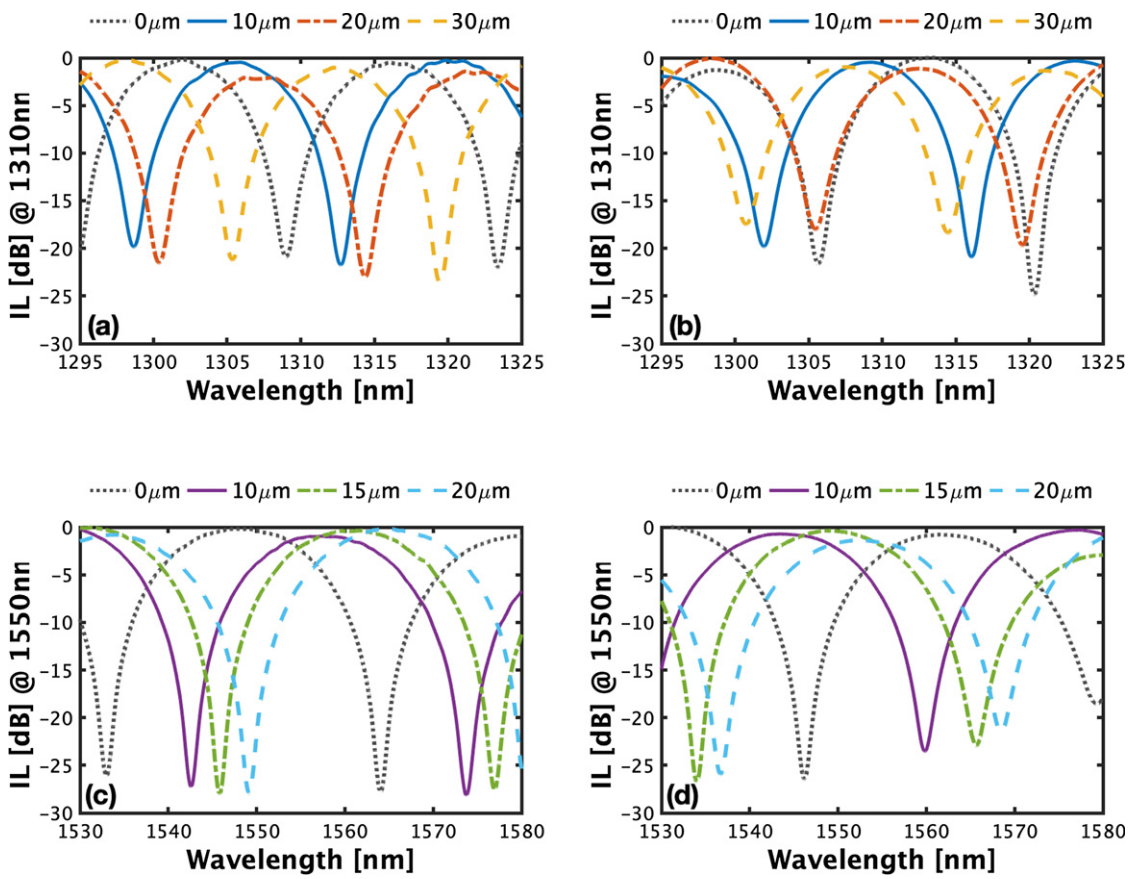

Figure 2. MZI response for different cell lengths, as indicated in the legends, of Sb_2S_3 in the (a) amorphous state at 1310 nm, (b) crystalline state at 1310 nm, (c) amorphous state at 1550 nm and (d) crystalline state at 1550 nm.

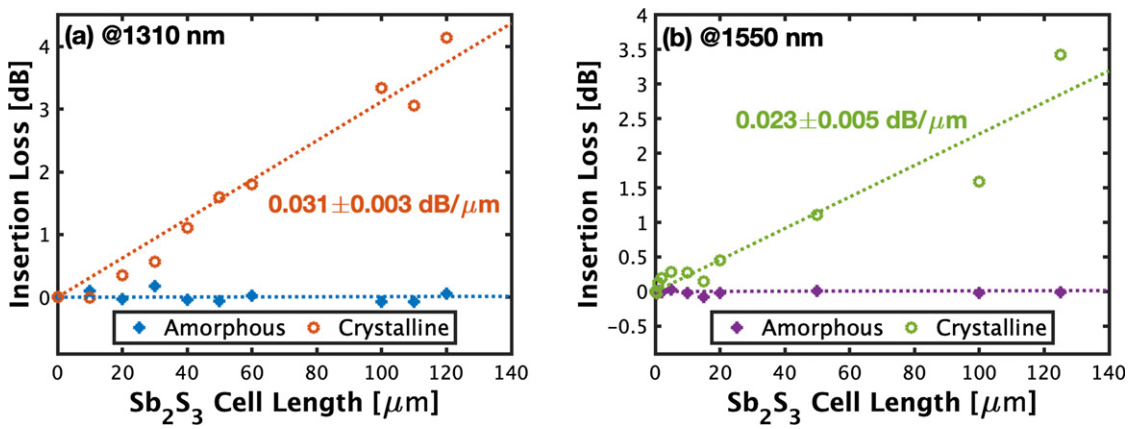

Figure 3. Measured losses for different cell lengths of Sb_2S_3 in both amorphous and crystalline state at (a) 1310 nm and (b) 1550 nm.

amorphous and crystalline states with respect to the dip produced by the bare MZI waveguide as introduced previously. Figure 4 shows the Δn_{eff} for both ranges of the spectrum, 1310 nm (figure 4(a)) and 1550 nm (figure 4(b)). The modulation between the amorphous state and the bare waveguide, the crystalline state and the bare waveguide and the amorphous state compared with the crystalline state (Δ) are presented in figure 4. The overall effective refractive index contrast between amorphous and crystalline states was measured to be (0.050±0.007) and (0.019±0.003) at 1310 nm and 1550 nm, respectively.

The same procedure was used in order to characterize the building block performance of the MZIs devices for the material, Sb_2Se_3. In this case, the spectral response for the different MZIs is presented in figure 5. Figure 5(a) shows the amorphous state of the Sb_2Se_3 material in the O-band. A shift of 9, 15.5 and 25.4 nm between the bare MZI dip and the MZI dip with PCM cells of 10, 20 and 30 μm was measured respectively. For the crystalline state, an increase in the shift is produced compared with the amorphous state and a shift of

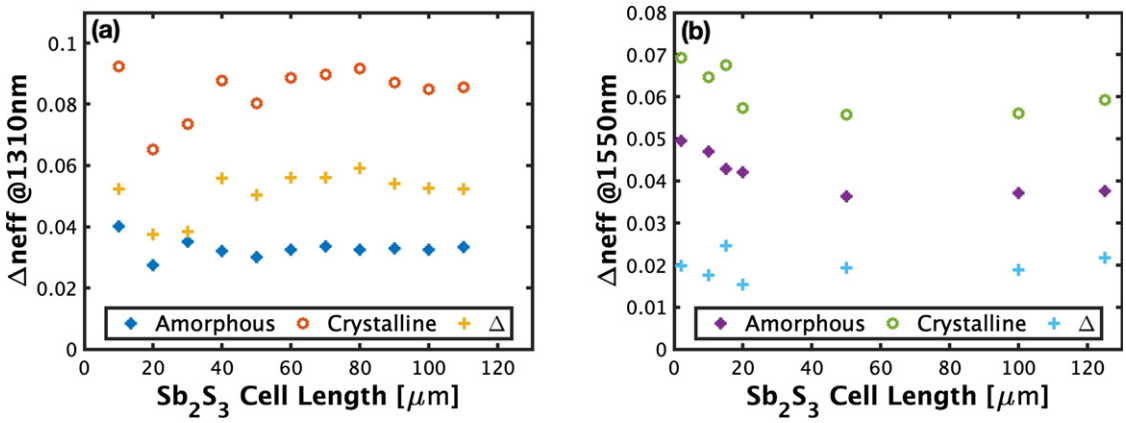

Figure 4. Effective refractive index difference ($\Delta_{n_{\mathrm{eff}}}$) as a function of cell length for both amorphous and crystalline states of Sb_2S_3 at (a) 1310 nm and (b) 1550 nm.

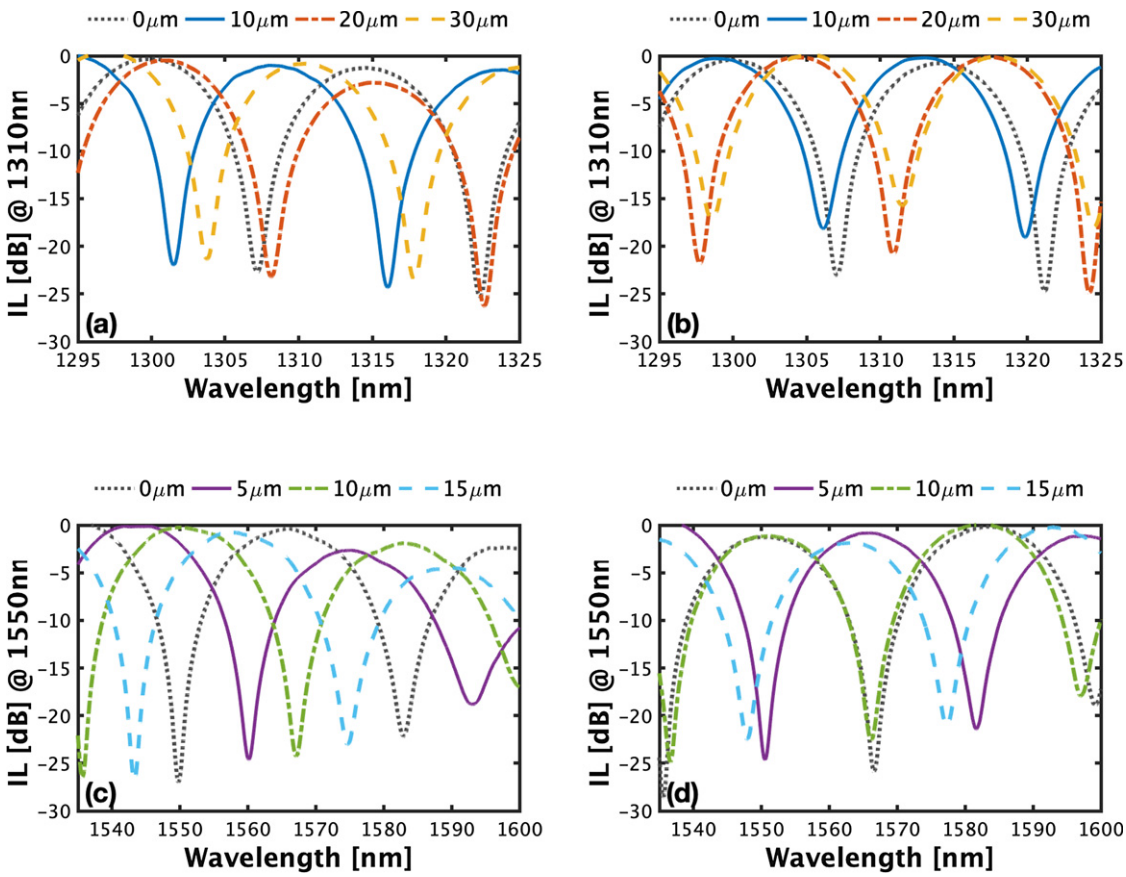

Figure 5. MZI response for different cell lengths, as indicated in the legends, of Sb_2Se_3 in the (a) amorphous state at 1310 nm, (b) crystalline state at 1310 nm, (c) amorphous state at 1550 nm and (d) crystalline state at 1550 nm.

13, 17.9 and 31.5 nm is measured for cell lengths of 10, 20 and 30 μm respectively. In all the cases ERs higher than 20 dB were demonstrated, see figure 5(b). In the C-band, the Sb_2Se_3 material showed a modulation in wavelength of 10.3, 17.5 and 24.9 nm for PCM amorphous cell lengths of 5, 10 and 15 μm. When switching to crystalline state, the modulation is 14.9, 30.65 and 41.4 nm respectively, see figures 5(c) and (d). As Sb_2Se_3 has a stronger modulation effect when compared to Sb_2S_3, smaller lengths of Sb_2Se_3 were chosen to be plotted to demonstrate an equivalent effect for both materials at the working wavelengths.

The ILs and the effective index modulation were extracted using the same method used for the Sb_2S_3 material. In this case, the measured losses in both states are shown in figure 6. In the amorphous state of Sb_2Se_3 for the wavelengths of interest, losses are lower than 10^{-4} dB μm^{-1}, whereas in the crystalline state, losses are (0.077±0.009) dB μm^{-1} at 1310 nm and (0.07±0.01) dB μm^{-1} at 1550 nm. The effective refractive index

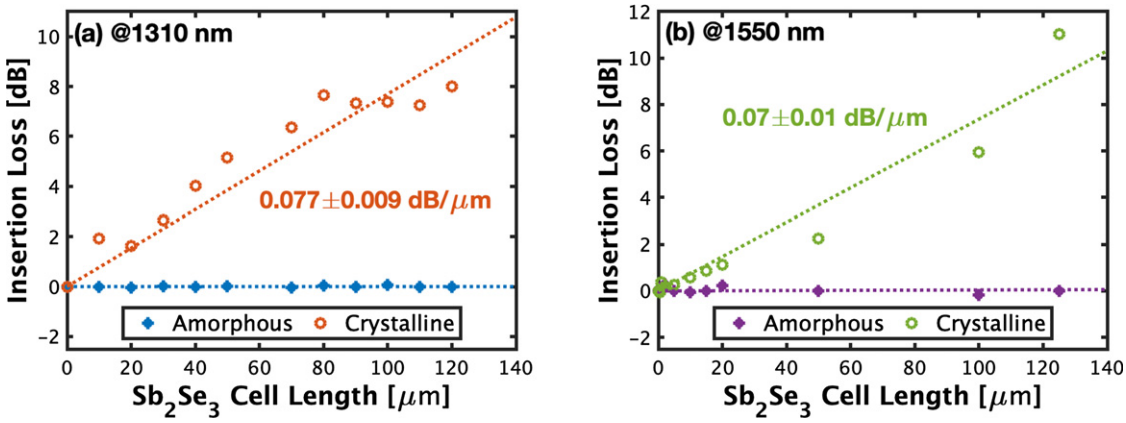

Figure 6. Measured losses for different cell lengths of Sb_2Se_3 in both amorphous and crystalline state at (a) 1310 nm and (b) 1550 nm.

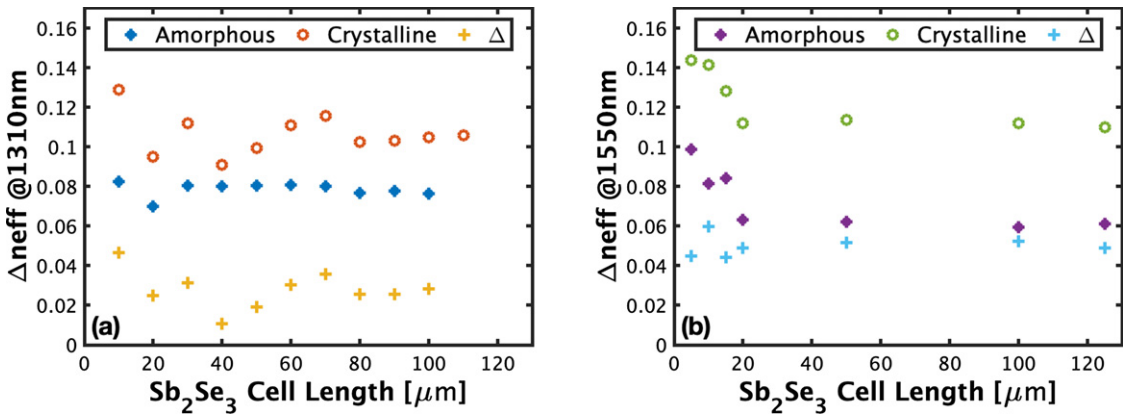

Figure 7. Effective refractive index difference ($\Delta_{n_{\text{eff}}}$) as a function of cell length for both amorphous and crystalline states of Sb_2Se_3 at (a) 1310 nm and (b) 1550 nm.

difference between the amorphous and crystalline state was measured to be (0.028±0.009) at 1310 nm and (0.050±0.005) at 1550 nm as shown in figure 7.

Table 1 presents the optical performance of the currently popular PCMs GST, GSST and the emerging materials discussed in this paper, Sb_2S_3 and Sb_2Se_3 on our integrated SiNx platform. In the table, a comparison of the cell length required in order to achieve a π shift ($L_\pi = \lambda/(2 \cdot \Delta n_{\text{eff}})$) and the IL that this cell introduces in the device are also shown. GST offers a π phase shift at the shortest length while GSST requires double, Sb_2Se_3 triple and Sb_2S_3 seven times that length (the GST length) for the same shift at 1550 nm. However, the popularity of GSST stems from the three times lower IL when compared to GST. Our Sb_2Se_3 and Sb_2S_3 films offer a stark improvement with the IL at 13 and 18 times lower than GST respectively, significantly surpassing the benefits of GSST. In the O-band (1310 nm), the required length for a π shift (L_π) in the case of Sb_2Se_3 is 21.83 μm, although, there is a reduction of 12.5 μm for GST, 14.5 μm for GSST and 9 μm for the Sb_2S_3 material. As with the C-band, the introduction of Sb_2S_3 and Sb_2Se_3 provides a significant performance increase in IL when compared to GST and GSST. It is therefore obvious that both materials provide in the O band and C band an incontestable improvement on ILs compared with GST and GSST in the crystalline state on a SiNx platform. The PCM thickness has a strong dependence in the IL and phase modulation. In the case of GST, going from 5 nm up to 15 nm thickness, an increase in the IL of 4 dB μm^{-1} occurs in the C-band and around 12.5 dB μm^{-1} in the O-band as can be seen in reference [57]. This is due to the stronger interaction of the mode with the PCM deposited on the waveguide surface and the higher refractive index that the PCM presents compared with the SiNx waveguide. In this table, we have used the thickness of GST and GSST, which the majority of published works utilized as the optimized one, 10 nm. Increasing the thickness of the GST and GSST layers will reduce the L_π but increase the ILs, which is the parameter that limits the number of cascading elements for neuromorphic computing. The required PCM length to achieve a π shift is related primarily to the effective refractive index contrast between the two phases of the material, but also to the waveguide geometry, operation wavelength and crystal size of the respective PCM. The crystal size has an impact in the phase modulation. As can be seen in

Table 1. The device performance comparison between the O-band (1310 nm) and the C-band (1550 nm) for different PCMs: GST [57], GSST [63], Sb_2S_3 and Sb_2Se_3. * refers to a simulated value.

PCM	O-band @ 1310 nm					C-band @ 1550 nm				
	α_c (dB μm^{-1})	α_a (dB μm^{-1})	Δn_{eff}	L_π (μm)	IL$_\pi$ (dB)	α_c (dB μm^{-1})	α_a (dB μm^{-1})	Δn_{eff}	L_π (μm)	IL$_\pi$ (dB)
GST [57]	6.660	0.228	0.07	9.36	62.4	2.860	0.039	0.14	5.53	15.81
GSST* [63]	1.42	$< 10^{-4}$	0.09	7.28	10.3	0.53	$< 10^{-4}$	0.074	10.5	5.56
Sb_2S_3	(0.031±0.003)	$< 10^{-4}$	(0.051±0.007)	13.1	0.39	(0.023±0.005)	$< 10^{-4}$	(0.019±0.003)	38.75	0.89
Sb_2Se_3	(0.077±0.009)	$< 10^{-4}$	(0.028±0.009)	21.83	1.68	(0.07±0.001)	$< 10^{-4}$	(0.050±0.005)	15.5	1.15

figures 4 and 7, the different crystal sizes of the PCMs produces a fluctuation in the crystalline measurement, which is reflected in the total phase modulation between the two states of the PCM. Different crystal sizes for a hot plate annealing between Sb_2S_3 and Sb_2Se_3 can be seen in [62]. Regarding the ILs of the PCM materials, the roughness scattering loss depends on the wavelength and for shorter wavelengths the roughness scattering increases as can be found in [69]. For that reason, the PCM materials in the O-band present higher losses compared to the C-band. The optical properties of the material in the crystalline state are not constant and heavily rely on the method and duration of heating but also on the switching history of the material. Effects such as vacancy ordering at different thermal energies can affect factors such as carrier mobility significantly affecting the optical constants [70]. In order to achieve similar refractive index modulation, much longer PCM cells are required for Sb_2Se_3 and Sb_2S_3 than for GST, however, GST results in higher ILs. The cells based on Sb_2Se_3 and Sb_2S_3 can be switched as efficiently as shorter GST cells, and this has been recently demonstrated on a silicon on insulator platform in [71].

The benefit of using SiN is extremely low losses, low temperature dependence and multilayer/BEOL capability. The PCM integrated to the SiN platform in this work demonstrates an improvement of 52% in the ER of the MZI building blocks compared with the SOI platform shown in [72]. Furthermore, SiN based platforms, due to the lower refractive index compared with SOI, requires smaller active regions to achieve the same phase modulation. In this work, an active region of 15.5 μm is required to achieve a π phase shift while as demonstrated in [72], 25 μm are required for obtaining the same phase modulation in SOI. Consequently, this work reduces by 62% the active region length compared with the SOI platform operating at 1550 nm and for the material Sb_2Se_3. The emerging materials Sb_2S_3 and Sb_2Se_3 can also be exploited in different regimes of the spectrum as demonstrated in [73] due to their large bandgap. Going to shorter wavelengths towards the visible range can be envisioned such that these materials are used in the optical absorption regime, similarly to GST in the NIR range which has been exploited for non-volatile optical memories [52]. To consider shorter wavelengths of operation (<400 nm), other waveguiding materials such as aluminium nitride can also be considered [74]. The benefits of using SiN are depicted for an individual building block which can be of benefit for highly integrated systems in neuromorphic computing where there is a need for non-volatile optical switches and optical memories to retain the weights associated to the learning of a neural network [75]. In these system, low optical losses and reduced energy consumption are key [38] and the importance and benefit of using a scalable complementary platform such as SiNx may prove valuable [37, 76–80].

4. Conclusions

In this paper, we demonstrate MZIs building blocks exploiting an emerging family of low loss PCMs in the O and C-telecommunications bands paving the way for future non-volatile electro-refractive modulation in mid index waveguides. Losses lower than 0.09 dB μm^{-1} for Sb_2Se_3 and 0.04 dB μm^{-1} for Sb_2S_3 are demonstrated in both ranges of the spectrum with changes in effective refractive index higher than 0.018. This study thus provides useful characterization information for the operation of non-volatile integrated photonic circuits based on PCMs in two of the most important spectral ranges used for optical communications combined with a SiNx platform. The demonstration of these low loss non volatile building blocks, using a low loss and back end of the line compatible waveguide platform, open a pathway for more complex scalable PICs and architectures for compute and routing applications.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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